1. Experimental Techniques and Chemical Analysis

1.1. Measurement

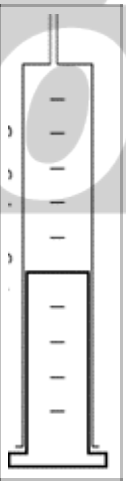
Variable	Unit(s)	Apparatus
Time	min/sec	Stopwatch
Temperature	°C	Thermometer
Mass	grams	Balance

Measuring Volume in Liquids

Approximate measure to most accurate measure.

Name	Measuring Cylinder	Volumetric Pipettes (fixed volumes)	Burettes (variable volume)
Image			

Measuring Volume in Gases

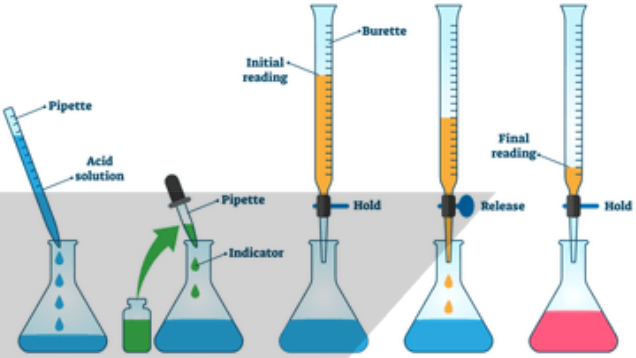
A gas syringe is used to measure the volume of gases gradually	

1.2. Mixture of Substances

- Mixture: A mixture is a substance made from two or more other substances not chemically bonded together
- Solution: A mixture of one or more solutes dissolved in a solvent
- Solute: A substance that dissolved in a solvent
- Solvent: A substance that dissolves a solute
- Saturated Solution: A solution containing the maximum amount of solute that can be dissolved in a solvent at a given temperature and pressure

1.3. Acid-Base Titration

A method of quantitative chemical analysis where an acid is added slowly to a base until it has been neutralised.



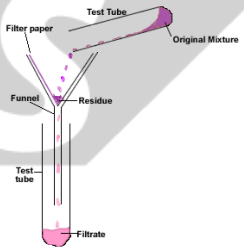
In Acid-Base Titrations, we must add a few drops of indicator (passes the endpoint) to check whether there are colour changes in the chemical reaction.

Indicators	Colour in Acid	Colour in Neutral	Colour in Base
Thymolphthalein	Colourless	Colourless	Blue
Methyl Orange	Red	Orange	Yellow

1.4. Filtration

Used to separate an insoluble solid from a liquid

- The mixture goes through a funnel with filter paper into a flask.
- Insoluble residue remains in the funnel
- Filtrate flows through the funnel and gets collected in the flask or test tube or a beaker



1.5. Crystallisation

Used to separate dissolved solid from a solution

- The solution is heated to increase concentration (solvent evaporates)
- A drop of solution is placed on a slide to check for crystal formation
- The solution is left till it reaches the crystallisation point.

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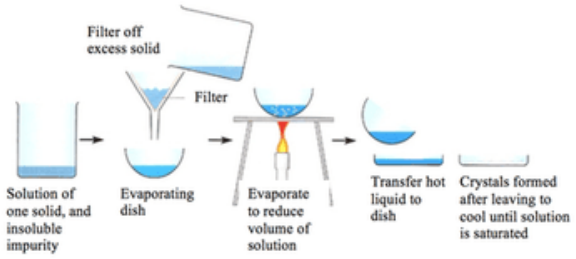
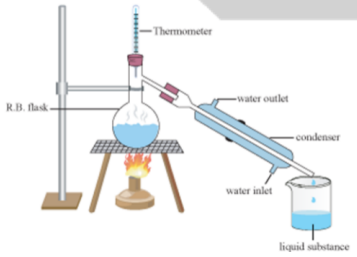
- Crystals are filtered from the solution, washed with distilled water, and dried between the filter paper.



1.6. Simple Distillation

Used to separate a solvent from a solution

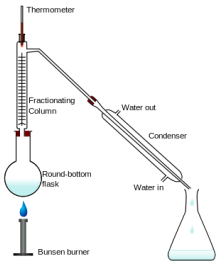
- The impure liquid is heated in a round bottom flask
- When it boils, the steam rises into the attached condenser
- Condenser cools the steam to a pure liquid, and it drops into the beaker
- A thermometer is placed to ensure the **highest boiling point is not exceeded.**



1.8. Fractional Distillation

Used to separate miscible liquids

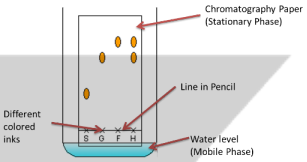
- Mixture is heated
- Substances, due to their different boiling points, rise in different fractions.
- A mixture of gases condenses on the beads in the fractional column.
- The beads are heated to the boiling point of the lowest substance so that the substance being removed cannot condense on the beads.
- The other substances continue to condense and will drip back into the flask.
- The beaker can be changed after every fraction



1.9. Chromatography

Used to separate substances in a solvent with different solubilities.

1. Drop the substance onto the start line (pencil) drawn on chromatography paper.
2. Paper is placed in a beaker with solvent; the paper must touch the surface of the solvent (**water or ethanol**) while the line must be above the liquid.
3. The solvent travels up the paper.
4. Different solubilities lead to different travel rates (**high solubility -> high travel rate**)
   - The stationary phase is the material on which separation takes place
   - The mobile phase is the mixture you want to separate, dissolved in a solvent.



- Interpreting simple chromatograms:

  Chromatograms are the visual outputs on the chromatography paper

  - **Number of rings/dots = number of substances**
  - If two dots travel the same distance up the paper, they are the **same substance.**
  - The pure substance only gives **one spot.**
- Retention Value:

  Used to identify a substance, calculated by the formula:

$$\text{Rf Value} = \frac{\text{Distance moved by substance}}{\text{Distance moved by solvent}}$$

Locating Agents

- Used to make colourless chromatograms visible
  - Dry paper in the oven
  - Spray it with locating agent
  - Heat it for 10 minutes in the oven

1.10. Separation Techniques

- This can be done by dissolving one in an appropriate solvent
- Then filter one and extract the other from the solution by evaporation
- If one solid is magnetic, it can use a magnet, e.g. sand and iron fillings

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Component 1	Component 2	Technique	Example
Solid	Solid	Use a solvent	Sand and Salt
Insoluble Solid	Liquid	Use Filtration - Separating the residue and the filtrate	Copper (II) Oxide and Water
Soluble Solid	Liquid	Crystallization - Liquid is evaporated to leave the solid	Copper (II) sulfate from water
Liquid	Liquid (Miscible)	Distillation (2 Liquids) or Fractional Distillation (More than two liquids)	Ethanol & Water (Simple), Crude Oil (Fractional Distillation)

1.11. Purification

Purity in Substances

- Assessing purity

Pure substances	Have a definite, sharp melting point/boiling point
Impure substances	Have a lower melting point and a higher boiling point

This assessment of substance purity is important, especially in food consumption, as its intake can be dangerous.

1.12. Identification of Ions and Gases

Test for Cations

Cations	With aqueous NaOH	With aqueous Ammonia
Aluminum (Al^{3+})	White soluble precipitate turns colourless in excess	White precipitate, insoluble in excess
Ammonium (NH_4^+)	Ammonia gas produced on warming	
Calcium (Ca^{2+})	White precipitate, insoluble in excess	Faint or no precipitate
Copper (Cu^{2+})	Light Blue precipitate, insoluble in excess	Light Blue precipitate, soluble in excess to give a dark blue solution
Iron(II) (Fe^{2+})	Green precipitate, insoluble in excess	Green precipitate, insoluble in excess
Iron(III) (Fe^{3+})	Red-brown precipitate, insoluble in excess	Red-brown precipitate, insoluble in excess
Zinc (Zn^{2+})	White precipitate, soluble and turns colourless in excess	White precipitate, soluble and turns colourless in excess
Chromium (Cr^{3+})	Green precipitate, soluble in excess	Green precipitate, insoluble in excess

Test for Anions

Anion	Test	Test Result
Carbonate (CO_3^{2-})	Add dilute acid, then test for carbon dioxide gas	Effervescence, carbon dioxide produced
Chloride (Cl^-)	Acidify with dilute nitric acid, then add aqueous silver nitrate	White ppt
Bromide (Br^-)	Acidify with dilute nitric acid, then add aqueous silver nitrate	Cream ppt
Iodide (I^-)	Acidify with dilute nitric acid, then add aqueous silver nitrate	Yellow ppt
Nitrate (NO_3^-)	Add aqueous sodium hydroxide, then aluminium foil; warm carefully	Ammonia produced
Sulfate (SO_4^{2-})	Acidify with dilute nitric acid, then add aqueous barium nitrate	White ppt
Sulfite (SO_3^{2-})	Add a small volume of acidified aqueous potassium manganate(VII)	The acidified aqueous potassium manganate(VII) changes colour from purple to colourless

Gas Tests

Gas	Test and Test Result
Ammonia (NH_3)	Damp red litmus paper turns blue
Carbon Dioxide (CO_2)	Turns limewater milky
Chlorine (Cl_2)	Bleaches red/blue litmus paper
Hydrogen (H_2)	Place lighted splint, squeaky pop
Oxygen (O_2)	Place glowing splint, splint relights
Sulfur Dioxide (SO_2)	Turns Acidified Aqueous Potassium Manganate (VII) from purple to colourless

Flame Tests

Metal Ion	Flame Colour
Lithium	Red
Sodium	Yellow
Calcium	Orange-red
Potassium	Lilac
Barium	Light-Green
Copper (II)	Blue-Green

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